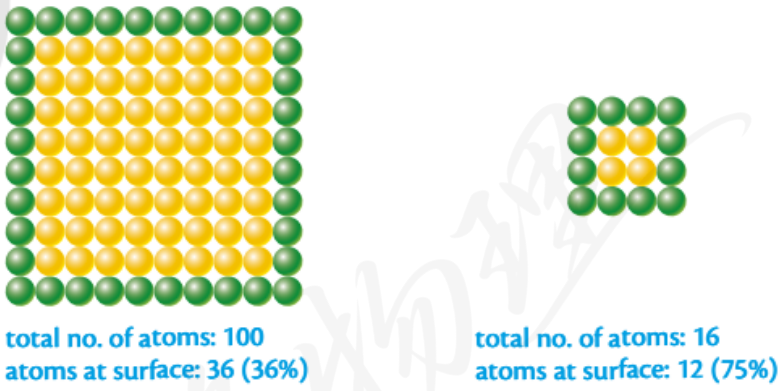
E2.3.1 Nanoscale materials

A nanometre is one billionth of a metre. Nanomaterials usually have at least one dimension around 1–100 nm. Increased surface-atom fractions and structural or quantum effects can change their electrical, mechanical and optical properties relative to bulk material. Do not attribute every change to surface area alone. Explain the material-specific mechanism.

$$1\ \text{nm} = 10^{-9}\ \text{m}$$

Nanomaterials typically have at least one dimension between 1 and 100 nm.



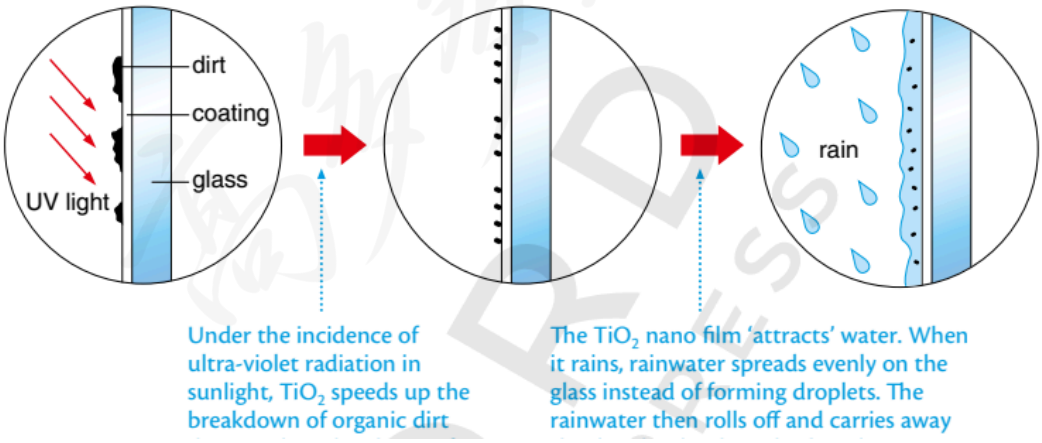
Smaller particles have a greater fraction of atoms at their surfaces.

E2.3.2 Self-cleaning coatings and carbon structures

Titanium-dioxide coatings clean in two ways. Ultraviolet light assists breakdown of organic dirt. The hydrophilic surface lets rain spread as a film and wash dirt away. A lotus leaf's hydrophobic surface instead forms droplets that carry dirt away as they roll.

Carbon materials show how atomic arrangement changes properties. The table compares four materials made from the same element but with different structures.

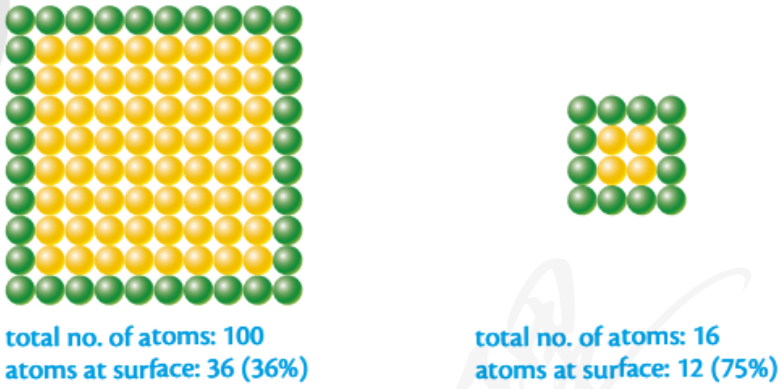
Material	Atomic arrangement	Related properties
Diamond	Three-dimensional covalent network	Hard; bound electrons make it normally insulating
Graphite	Stacked hexagonal layers with weaker forces between layers	Layers slide; conducts within layers
Graphene	One honeycomb carbon sheet	Thin, strong and conducting
Carbon nanotube	Tubular carbon network	Strong; electrical behaviour depends on tube structure



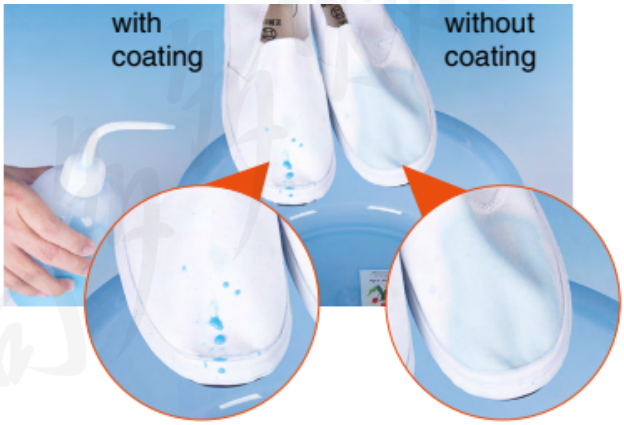
Hydrophilic self-cleaning by a titanium-dioxide coating.

E2.3.3 Surface effects and the lotus effect

Reducing size increases surface-area-to-volume ratio and the fraction of surface atoms available to interact with the surroundings. A lotus leaf's nanoscale surface structures repel water, allowing rolling droplets to carry dirt away.



Smaller particles have a greater fraction of atoms at their surfaces.



The water-repelling lotus surface lets droplets roll away carrying dirt.

E2.3.4 Rayleigh's criterion and resolving power

A finite aperture turns each point source's image into a diffraction pattern with a central bright spot and surrounding bright and dark rings. Nearby point sources produce overlapping patterns.

Rayleigh's criterion defines just-resolved images when one central maximum falls on the other's first dark ring. For a circular aperture, the minimum resolvable angle θ_R increases with wavelength λ and decreases with diameter d. Shorter wavelength or a larger aperture resolves closer points. Magnifying an image and separating overlapping point images are different functions.

$$\theta_R \approx 1.22\frac{\lambda}{d}$$

Rayleigh criterion for a circular aperture. θ_R: minimum resolvable angular separation (rad); λ: wavelength (m); d: aperture diameter (m), or pupil diameter for the eye.

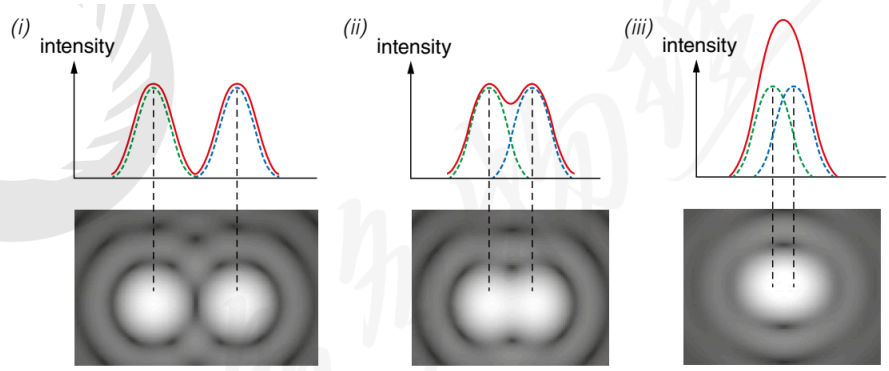


Fig 3.2d *According to Rayleigh criterion, two objects are (i) resolved clearly, (ii) just resolved, and (iii) not resolved.*

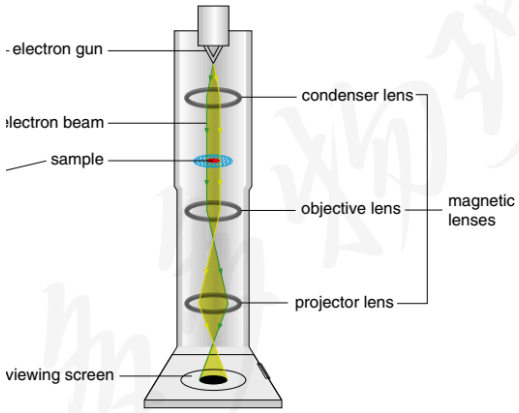
Compare clearly resolved, just resolved and unresolved pairs. At the Rayleigh limit, one central maximum falls on the first minimum of the other pattern.

E2.3.5 Components of a transmission electron microscope

A transmission electron microscope first uses an electron gun to produce a beam. Condenser magnetic lenses focus it onto a thin sample. Transmitted electrons pass through objective and projector lenses to form an enlarged image, recorded on a screen or camera. Different parts of the sample transmit electrons differently, creating contrast. Magnetic lenses change electron directions using magnetic fields.

Transmission electron microscopy

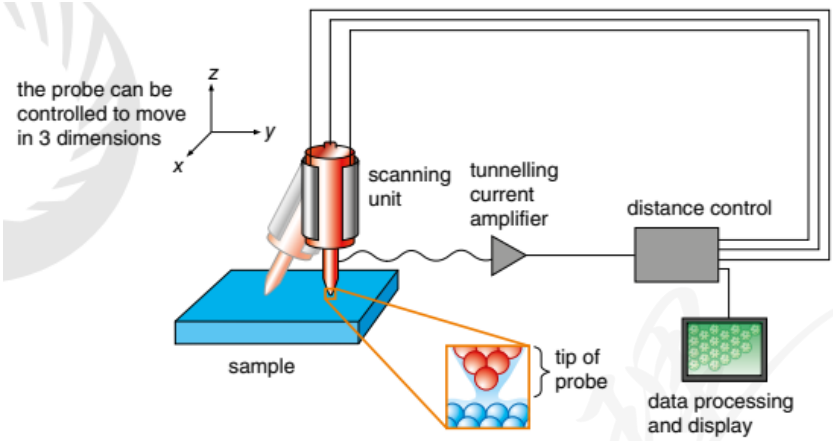
A transmission electron microscope passes fast electrons through a thin sample. Transmission differences create contrast, and magnetic lenses focus and magnify the beam. Higher accelerating voltage increases momentum and reduces de Broglie wavelength, improving the wavelength-related resolution limit. Actual resolution is also limited by lens and specimen effects.



Follow the electron beam through the gun, magnetic lenses and sample to the imaging screen.

E2.3.6 Scanning tunnelling microscopy

A conducting tip near a conducting sample detects a tunnelling current highly sensitive to separation. Scanning at fixed height while recording current, or adjusting height to keep current constant, reveals surface detail. The image is reconstructed from scanning and tunnelling signals.



A scanning tip detects tunnelling current across a very small gap to obtain surface information.